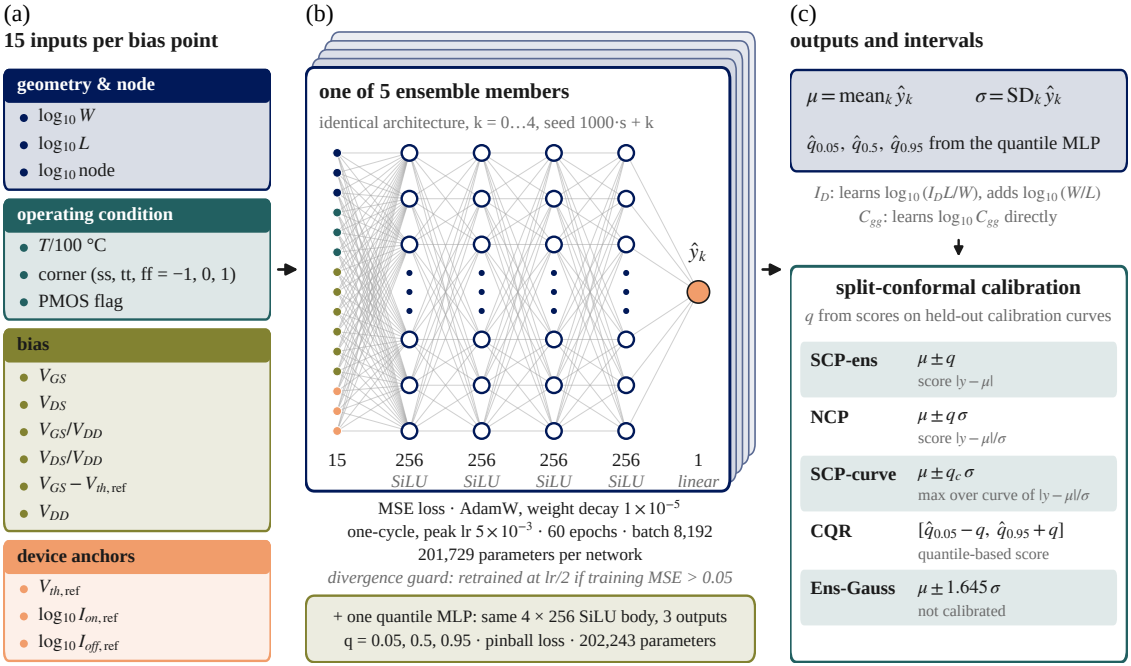

(a)
15 inputs per bias point
geometry & node
$\log_{10} W$
$\log_{10} L$
$\log_{10}$ node
operating condition
T/100 °C
corner (ss, tt, ff = −1, 0, 1)
PMOS flag
bias
V_{GS}
V_{DS}
V_{GS}/V_{DD}
V_{DS}/V_{DD}
$V_{GS} - V_{th,\mathrm{ref}}$
V_{DD}
device anchors
$V_{th,\mathrm{ref}}$
$\log_{10} I_{on,\mathrm{ref}}$
$\log_{10} I_{off,\mathrm{ref}}$
(b)
one of 5 ensemble members
identical architecture, k = 0…4, seed 1000·s + k
$\hat{y}_k$
15
256 SiLU
256 SiLU
256 SiLU
256 SiLU
1 linear
MSE loss · AdamW, weight decay 1×10^{-5}
one-cycle, peak lr 5×10^{-3} · 60 epochs · batch 8,192
201,729 parameters per network
divergence guard: retrained at lr/2 if training MSE > 0.05
+ one quantile MLP: same 4 × 256 SiLU body, 3 outputs
q = 0.05, 0.5, 0.95 · pinball loss · 202,243 parameters
(c)
outputs and intervals
$\mu = \mathrm{mean}_k\, \hat{y}_k$ $\sigma = \mathrm{SD}_k\, \hat{y}_k$
$\hat{q}_{0.05}$, $\hat{q}_{0.5}$, $\hat{q}_{0.95}$ from the quantile MLP
I_D: learns $\log_{10}(I_D L/W)$, adds $\log_{10}(W/L)$
C_{gg}: learns $\log_{10} C_{gg}$ directly
split-conformal calibration
q from scores on held-out calibration curves
SCP-ens $\mu \pm q$ score $|y - \mu|$
NCP $\mu \pm q\,\sigma$ score $|y - \mu|/\sigma$
SCP-curve $\mu \pm q_c\,\sigma$ max over curve of $|y - \mu|/\sigma$
CQR $[\hat{q}_{0.05} - q,\ \hat{q}_{0.95} + q]$ quantile-based score
Ens-Gauss $\mu \pm 1.645\,\sigma$ not calibrated